

Identification and Classification of Encrypted Network Traffic Flows

Tony Vukovic
202512471

Arij Kristou
202512443

Aurore Bellessa
202511831

Introduction

The rapid proliferation of encrypted network protocols, including TLS 1.3, QUIC, and DNS-over-HTTPS (DoH), has fundamentally altered the threat detection landscape. As encryption becomes the default standard for internet communications [3], the effectiveness of traditional Deep Packet Inspection (DPI) techniques is significantly diminished : security analysts can no longer rely on payload content to identify malicious activity [1].

In response, the research community has shifted toward machine learning (ML) approaches that operate exclusively on flow-level metadata, statistical descriptors such as inter-arrival times, packet length distributions, and TCP window initialization sequences, without requiring decryption [2]. This paradigm offers a critical advantage : it preserves user privacy while maintaining the capacity to detect threats. However, it also introduces two interdependent challenges. First, it remains an open question which flow features most reliably indicate malicious behavior within a high-entropy, encrypted environment. Second, the tools used to extract such features vary considerably in the richness and nature of the metadata they produce, making tool selection a non-trivial decision for practitioners.

This work addresses both challenges. We investigate the feasibility of classifying threats in encrypted network traffic using only flow-level statistical metadata, with no access to packet payloads, IP addresses, or port numbers. To this end, we apply a behavioral fingerprinting strategy to isolate encrypted flows from the CIC-IDS benchmark dataset collection and evaluate the performance of a machine learning classifier across several attack categories including DoS, DDoS, Botnet, Brute Force, and Infiltration. In parallel, we examine and compare existing flow extraction frameworks in terms of the metadata coverage they offer for encrypted traffic analysis.

This paper asks : to what extent can statistical flow metadata alone support reliable, privacy-preserving threat detection in fully encrypted network environments, and how does the choice of extraction tooling affect that capacity? The remainder of this paper presents our methodology, experimental results, and a discussion of the limits of decryption-free classi-

fication. You may find the link to access the code here : <https://github.com/arijkrisout-hash/Security-of-system-and-networks>

I State of the Art in Encrypted Traffic Feature Engineering

I.1 Encrypted Traffic Classification Landscape

ML based network traffic classification is a well-established field in network security. Early foundational work [4, 5] demonstrated that statistical flow metrics alone, without payload access, suffice for highly accurate application identification. As internet privacy protocols evolved, more recent work has successfully extended these statistical methodologies to fully encrypted traffic environments. Researchers now heavily rely on transport-layer characteristics, such as TLS fingerprinting via client-server JA3 and JA3S hashes, QUIC protocol metadata, and complex inter-packet timing (IPT) or packet length distributions, to identify applications and detect anomalous behaviors.

I.2 Feature Extraction Tools

To capture these structural characteristics, practitioners utilize various open-source flow processing frameworks, though each offers distinct architectural trade-offs. *NFStream* stands out as a Python-native framework optimized for extracting rich, highly customizable flow-level statistics [6]. In contrast, *Cisco Joy* focuses heavily on security-oriented telemetry, specifically TLS/SSH fingerprinting and cryptographic handshake metrics. For high-throughput environments, *PcapPlusPlus* offers optimized C++ performance designed for high-speed packet capture and multi-core processing. Additionally, developer libraries like *PyShark* and *Scapy* are widely used within the research community as ground-truth verification tools for granular packet-by-packet validation.

I.3 Benchmark Datasets

The evaluation of contemporary intrusion detection systems relies heavily on standard reference datasets, like the Canadian Institute for Cybersecurity family, including CIC-IDS2017, CSE-CIC-IDS2018, and CIC-DDoS2019. A well-documented, inherent limitation of these datasets is the severe class imbalance between the massive volume of benign baseline traffic and the rare occurrences of attack flows. This disproportionate distribution directly impacts machine learning performance, particularly on stealthy, multi-stage vectors like the Infiltration class. To mitigate these dataset-specific biases, our study utilizes the harmonized *CIC-IDS-Collection*, a consolidated framework available on Kaggle that standardizes features and labels across these disparate datasets to enable a robust, comprehensive multi-attack evaluation.

II Project Contribution

II.1 Dataset and Encrypted Traffic Isolation

This study leverages the harmonized CIC-IDS-Collection, a consolidated benchmark repository unifying CIC-IDS2017, CIC-DDoS2017, CSE-CIC-IDS2018, and CIC-DDoS2019 into a single standardized framework of 9,167,581 flow records across 59 statistical features. To restrict the analysis exclusively to encrypted traffic, a behavioral fingerprinting heuristic is applied in the absence of port identifiers, which were deliberately excluded to prevent overfitting. Network flows are retained only when they exhibit strictly positive initial window sizes in both directions, captured by `Init Fwd Win Bytes` and `Init Bwd Win Bytes`, combined with a minimum forward segment length of 20 bytes (`Fwd Seg Size Min` ≥ 20). This multi-condition filter isolates the bidirectional handshake patterns characteristic of TLS and SSH sessions, yielding a reduced corpus of 4,758,611 encrypted flows distributed across eight attack categories: Benign, DDoS, DoS, Botnet, Bruteforce, Infiltration, Webattack, and Portscan.

II.2 Feature Engineering

Beyond the 57 native flow statistics provided by the dataset, a custom feature, `Flow_Repetition_Count`, is introduced. It is computed as the number of flows sharing identical mean and maximum forward packet length values, thereby capturing the repetitive volumetric signature of distributed attacks originating from multiple sources. All features are subsequently normalized using `StandardScaler` prior to model training. It is worth noting that this custom feature does not appear among the 15 most important features as ranked by the Random Forest, which itself constitutes an informative negative result regarding the discriminative power of repetition-based behavioral signatures in this setting.

II.3 Classification Pipeline and Iterative Refinement

A Random Forest classifier (50 estimators, max depth 15, stratified 70/30 split, `StandardScaler` normalization) constitutes the baseline model, trained on the full encrypted subset with no sampling correction. Given the baseline model’s difficulty with the Infiltration class, two complementary strategies are subsequently explored.

First, a Multi-Layer Perceptron (2 hidden layers of 50 and 25 neurons, early stopping, max 30 iterations) is trained to investigate whether a non-linear architecture could better capture the decision boundary between stealthy and benign flows. Second, two resampling strategies are evaluated to address class imbalance: SMOTE oversampling targeting minority classes, and a cost-sensitive Random Forest using `class_weight='balanced'` that deliberately prioritizes threat detection sensitivity over global accuracy. The results of this iterative refinement are discussed in Section III.2.

II.4 Tool Comparison

In parallel with the classification pipeline, three principal flow extraction frameworks — `NFStream`, `Cisco Joy`, and `PcapPlusPlus` — are evaluated with respect to the metadata coverage they offer for encrypted traffic analysis. Each framework presents a distinct architectural philosophy that directly conditions the nature and richness of the features it produces.

`NFStream` is a Python-native framework designed for offline and live flow-level statistical extraction, producing a broad set of per-flow metrics that align naturally with ML-based classification pipelines. `Cisco Joy` is a security-oriented tool focused on cryptographic telemetry, capturing TLS and SSH handshake properties that pure flow statistics cannot expose, but operating at a coarser binary granularity. `PcapPlusPlus` is a high-performance C++ engine optimized for high-speed packet capture, prioritizing throughput over statistical feature richness.

The practical implications of these architectural differences for encrypted traffic classification are evaluated empirically in Section III.3, where the same Random Forest pipeline is applied independently to each tool’s output.

II.5 Perspectives

As a natural extension of the offline pipeline toward operational deployment, an integration of the trained Random Forest model with `NFStream`’s live streaming API is prototyped, targeting real-time per-flow classification on a local network interface. This proof-of-concept represents the primary axis for future work, bridging the gap between controlled benchmark evaluation and production-grade encrypted traffic monitoring.

III Assessment

III.1 Feature Importance

Feature importance analysis on the baseline Random Forest reveals that the classifier relies predominantly on packet geometry rather than aggregate flow volume. As shown in Table 2, backward packet length statistics (*Bwd Packet Length Std*, *Mean*, and *Max*) account for approximately 17.7% of total feature weight, occupying the top three positions. This reflects a fundamental asymmetry in encrypted sessions : client-side requests are typically short and uniform, while server responses vary significantly depending on the service or attack type. Backward flows therefore expose a more consistent behavioral fingerprint than forward flows.

The high ranking of *Packet Length Std* (5th) and *Packet Length Variance* (9th) in Table 2 reinforces this finding. While encryption conceals payload content, it does not eliminate the structural regularity with which attack tools size their packets — automated malicious behavior tends to operate at fixed size regimes, a side-channel that persists under encryption. Cumulative volume metrics rank considerably lower, confirming that packet shape is a more robust discriminator than total bytes transferred.

Finally, the custom *Flow_Repetition_Count* feature did not appear among the top 15 features of Table 2, suggesting that repetition-based signatures do not contribute discriminative signal beyond what packet geometry already captures.

III.2 Machine Learning Model Comparison

The evaluation of the classification pipeline focused on comparing several learning strategies, whose detailed results are reported in Table 1. The baseline Random Forest achieved near-perfect precision and recall on large-scale attack categories such as DDoS, DoS, Botnet, and Bruteforce, confirming that encrypted traffic exposes sufficiently strong statistical side-channels for volumetric threat detection. However, it struggled significantly on the Infiltration class, revealing a substantial feature-space overlap between stealthy malicious flows and legitimate encrypted sessions.

To address the severe class imbalance, SMOTE oversampling was applied to synthetically generate additional minority-class samples. As shown in Table 1, this configuration preserved the 99% global accuracy of the baseline while marginally improving Infiltration precision to 0.60, yet recall remained critically low at 0.03, yielding an F1-score of only 0.06. This result indicates that the Infiltration detection difficulty originates not from class imbalance alone, but from the intrinsic similarity between infiltration flows and benign encrypted traffic in the selected feature space.

To investigate the opposite trade-off, a cost-sensitive Random Forest using `class_weight='balanced'` was evaluated, deliberately prioritizing minority-class sensitivity. Table 1 shows that Infiltration recall in-

creased dramatically to 0.93, but precision collapsed to 0.05, producing an extremely high false-positive rate and reducing global accuracy to 82.51%. This highlights the fundamental tension in intrusion detection between sensitivity and operational reliability.

Finally, a Multi-Layer Perceptron (2 hidden layers of 50 and 25 neurons) was trained to investigate whether a non-linear architecture could better separate stealthy attacks from benign flows. As reported in Table 1, the MLP achieved 98.45% global accuracy but yielded an Infiltration F1-score of only 0.13, offering no meaningful improvement over the baseline. This confirms that the limitation stems from the available behavioral representation rather than the choice of learning algorithm, and that richer protocol-level telemetry or temporal modeling may be necessary for robust infiltration detection.

III.3 Flow Extraction Tool Comparison

To evaluate the impact of feature extraction tooling on classification performance, the same Random Forest pipeline was applied independently to features extracted by NFStream, PcapPlusPlus, and Cisco Joy. The results reveal that tool selection is not a neutral decision — each framework introduces structural constraints that fundamentally shape what the classifier can and cannot learn.

NFStream achieved 91% global accuracy, with strong performance on Benign ($F1 = 0.99$) and DDoS ($F1 = 0.75$) traffic, while all remaining attack classes scored an $F1$ of 0.00 (see Table 3 in Appendix). A notable exception is the Infiltration class, which achieved a perfect $F1 = 1.00$. However, this result is attributed to feature leakage rather than genuine classification capability : as shown in Table 4, the two dominant features, *application_is_guessed* (0.51) and *application_confidence* (0.41), together account for over 92% of the model’s decision weight. NFStream’s built-in application detection engine likely encodes class-discriminative information directly into these features, making the Infiltration task trivially separable rather than genuinely learned. This fundamentally departs from the privacy-preserving, payload-free classification objective of this study, and would fail in environments where application-layer inference is unavailable or unreliable. The remaining flow-level features contribute negligibly, suggesting that NFStream’s rich statistical output is effectively overshadowed when application metadata is present.

PcapPlusPlus achieved 78% global accuracy, driven entirely by the majority Benign class ($F1 = 0.88$), with all attack categories scoring $F1 = 0.00$ (see Table 5 in Appendix). The feature importance analysis exposes the root cause : only two features were extracted — *has_syn_flag* and *packet_count* — both with an importance ratio of 0.0000 (see Table 6 in Appendix). This result is not a failure of the classifier but a direct consequence of PcapPlusPlus’s architectural focus on high-speed packet capture rather than offline statistical flow characterization. The framework is not desi-

igned to produce the rich per-flow metadata required for ML-based threat detection, and this experiment quantitatively confirms that limitation.

Cisco Joy achieved 79.44% global accuracy under both standard and algorithmically balanced training configurations, with the model almost entirely predicting Benign traffic (recall = 1.00) and nearly ignoring the Attack class (recall = 0.05, F1 = 0.09), as reported in Table 7 in Appendix. Notably, Cisco Joy’s binary Attack/Benign schema prevents any per-category analysis, which is itself an informative limitation : the tool’s security-oriented telemetry is designed around coarse anomaly flagging rather than fine-grained multi-class discrimination. The identical performance across both training configurations further suggests that the available features lack the discriminative structure needed to separate attack from benign flows, regardless of class balancing strategy.

Taken together, these results demonstrate that no single tool provides a complete solution for encrypted traffic classification. NFStream offers the richest statistical feature space but introduces application-layer metadata that can dominate and distort purely flow-based classifiers. PcapPlusPlus excels in throughput performance but is unsuitable for offline statistical analysis due to its limited feature output. Cisco Joy provides valuable cryptographic telemetry but restricts classification granularity to binary outcomes. A robust production pipeline would therefore benefit from a multi-tool approach, combining NFStream’s statistical depth with Cisco Joy’s handshake fingerprinting, while reserving PcapPlusPlus for high-speed capture contexts.

Conclusion

This study investigated the feasibility of classifying encrypted network traffic threats using exclusively flow-level statistical metadata, without access to packet payloads, IP addresses, or port numbers. The results demonstrate that this approach is highly effective for volumetric attack categories — DDoS, DoS, Botnet, and Bruteforce — where packet geometry and server-response sizing provide strong, persistent side-channels that survive encryption. However, the approach reaches a fundamental limit with stealthy, low-volume attacks such as Infiltration, where the available behavioral representation overlaps substantially with legitimate encrypted sessions. Neither synthetic oversampling, cost-sensitive learning, nor increased model complexity resolves this limitation, suggesting that richer protocol-level telemetry or temporal behavioral modeling would be necessary for robust detection of such threats.

The tool comparison further reveals that feature extraction tooling is a non-trivial architectural decision. NFStream, PcapPlusPlus, and Cisco Joy each introduce structural constraints that directly condition classification capability, and no single framework covers all dimensions of encrypted flow characterization. A multi-tool pipeline combining NFStream’s statistical depth with Cisco Joy’s cryptographic telemetry represents the most promising direction for future work, alongside the integration of the trained classifier into a live traffic monitoring system via NFStream’s streaming API.

Bibliographie

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Glossary of Technical Terms and Definitions

This appendix provides precise technical definitions for the protocols, cryptographic mechanisms, extraction frameworks, and benchmark datasets evaluated and referenced throughout this study.

Core Cryptographic and Network Protocols

TLS 1.3 (Transport Layer Security, Version 1.3)

The standard cryptographic protocol used to secure internet communications. Version 1.3 optimizes performance and enhances privacy by encrypting the majority of the handshake process (including server certificates) immediately following the initial cleartext client hello message, significantly reducing plaintext metadata compared to older iterations.

QUIC (Quick UDP Internet Connections)

A transport-layer network protocol running over UDP instead of TCP. It natively integrates cryptographic handshakes modeled on TLS 1.3 logic to achieve low-latency, zero round-trip connection establishment, while effectively obscuring traditional layer-4 transport control indicators from middleboxes.

DoH (DNS over HTTPS)

A protocol designed to perform remote Domain Name System (DNS) resolution using the encrypted HTTPS protocol via port 443. By wrapping traditional cleartext DNS queries in standard web traffic, it prevents eavesdropping and passive monitoring of user domain resolutions.

Cryptographic Fingerprinting Mechanisms

JA3 Fingerprinting

An open-source method utilized to profile client applications during the TLS handshake. It extracts explicit cleartext parameters from the initial ‘Client Hello’ packet—such as version, accepted ciphers, and extensions—and concatenates them into a single string to generate a unique MD5 signature representing the client software.

JA3S Fingerprinting

The server-side counterpart to the JA3 algorithm. It logs and hashes the parameters of the ‘Server Hello’ response packet, providing defenders with a structural behavioral signature of the server-side cryptographic architecture and assisting in the identification of command-and-control infrastructure.

Statistical Side-Channels and Flow Metrics

Flow-Level Metadata

Aggregate statistical measurements summarizing a distinct network session (defined by a standard 5-tuple : source IP, destination IP, source port, destination port, and protocol). It abstracts traffic behaviors based on data flow characteristics rather than inspecting raw packet payloads.

Inter-Packet Timing (IPT) / Inter-Arrival Time (IAT)

The discrete temporal interval between consecutive packet arrivals within a single network stream. IPT distributions serve as a powerful behavioral proxy, as machine-driven activity (e.g., automated beaconing) exhibits unique temporal frequencies compared to human web interactions.

Packet Length Distributions

A statistical distribution summarizing the various physical sizes (in bytes) of packets transmitted across a connection. It captures volumetric traffic trends and payload sizing behaviors that persist despite packet content encryption.

TCP Window Initialization Sequence

The initial window sizing and configuration parameters exchanged during the TCP three-way handshake. These base metrics differ across operating systems and network stacks, providing additional passive identifiers for machine learning models.

Feature Extraction Frameworks and Libraries

NFStream

A highly extensible, Python-native network data analysis framework designed for fast processing of live or offline traffic. It is specifically optimized to extract deeply structured flow-level metadata and time-series statistics.

Cisco Joy

An open-source data collection and parsing tool created to capture and evaluate security-oriented network telemetry, focusing explicitly on TLS/SSH handshakes and cryptographic properties.

PcapPlusPlus

A high-performance C++ wrapper and engine built for multi-platform packet capturing, dissecting, and crafting. It provides native multi-core optimization suited for high-speed network engineering workloads.

PyShark

A Python wrapper module built on top of the TShark (Wireshark) terminal packet dissector, enabling deep, granular packet-by-packet debugging and programmatic analysis.

Scapy

A versatile, interactive Python-based packet manipulation utility capable of forging, sniffing, decoding, and sending raw packets across a massive array of network protocols.

Benchmark Datasets**CIC-IDS Family**

A collection of foundational intrusion detection benchmark datasets engineered by the Canadian Institute for Cybersecurity (including CIC-IDS2017, CSE-CIC-IDS2018, and CIC-DDoS2019) containing modeled benign traffic baseline mixes alongside contemporary network threat vectors.

CIC-IDS-Collection

A unified, harmonized iteration of the standard CIC datasets hosted on Kaggle that maps and standardizes disparate labels, class identifiers, and underlying features to facilitate standardized, cross-dataset machine learning testing.

Infiltration Class

A highly specialized, multi-stage attack category represented within the CIC datasets that models external adversaries gaining local network footholds. It presents an acute optimization bottleneck for ML classifiers due to its stealthy footprint and severe class imbalance.

Performance Tables

TABLE 1 – Performance Metrics for Infiltration Detection across Models

Model	Accuracy	Inf. Recall	Inf. Precision	Inf. F1
RF + Class Weight	82.51%	0.93	0.05	0.09
RF + SMOTE	99.00%	0.03	0.60	0.06
MLP Neural Network	98.45%	0.11	0.17	0.13

TABLE 2 – Top 15 Weighted Features from Random Forest Analysis

Feature	Importance
Bwd Packet Length Std	0.0678
Bwd Packet Length Mean	0.0591
Bwd Packet Length Max	0.0508
Subflow Fwd Packets	0.0505
Packet Length Std	0.0460
Fwd Header Length	0.0445
Bwd Packets Length Total	0.0444
Total Fwd Packets	0.0444
Packet Length Variance	0.0370
Packet Length Mean	0.0342
Fwd Packets Length Total	0.0315
Subflow Fwd Bytes	0.0303
Fwd IAT Total	0.0267
Avg Bwd Segment Size	0.0261
Avg Fwd Segment Size	0.0248

TABLE 3 – NFStream — Classification Report

Class	Precision	Recall	F1-Score	Support
Benign	0.97	1.00	0.99	23516
Botnet	0.00	0.00	0.00	478
Bruteforce	0.00	0.00	0.00	338
DDoS	0.65	0.89	0.75	4040
DoS	0.00	0.00	0.00	1300
Infiltration	1.00	1.00	1.00	310
Portscan	0.00	0.00	0.00	7
Webattack	0.00	0.00	0.00	10
Accuracy			0.91	29999
Macro avg	0.33	0.36	0.34	29999
Weighted avg	0.86	0.91	0.88	29999

TABLE 4 – NFStream — Top 10 Feature Importances

Rank	Feature	Importance
1	application_is_guessed	0.5133
2	application_confidence	0.4069
3	src2dst_bytes	0.0141
4	bidirectional_bytes	0.0119
5	bidirectional_max_ps	0.0111
6	src2dst_min_ps	0.0110
7	src2dst_max_ps	0.0089
8	bidirectional_min_ps	0.0083
9	bidirectional_mean_ps	0.0082
10	src2dst_mean_ps	0.0063

TABLE 5 – PcapPlusPlus — Classification Report

Class	Precision	Recall	F1-Score	Support
Benign	0.78	1.00	0.88	23516
Botnet	0.00	0.00	0.00	478
Bruteforce	0.00	0.00	0.00	338
DDoS	0.00	0.00	0.00	4040
DoS	0.00	0.00	0.00	1300
Infiltration	0.00	0.00	0.00	310
Portscan	0.00	0.00	0.00	7
Webattack	0.00	0.00	0.00	10
Accuracy			0.78	29999
Macro avg	0.10	0.12	0.11	29999
Weighted avg	0.61	0.78	0.69	29999

TABLE 6 – PcapPlusPlus — Feature Importances

Rank	Feature	Importance
1	has_syn_flag	0.0000
2	packet_count	0.0000

TABLE 7 – Cisco Joy — Classification Report

Class	Precision	Recall	F1-Score	Support
Attack	1.00	0.05	0.09	4322
Benign	0.79	1.00	0.88	15678
Accuracy			0.79	20000
Macro avg	0.90	0.52	0.49	20000
Weighted avg	0.84	0.79	0.71	20000